HW 8.

Q.2:

Steel

$h = 25\ W/m^2K$

$k = 40\ W/m.K$

$\rho = 7800\ kg/m^3$

$c = 600\ J/kg.K$

$r = 5mm = 0.005\ m$

$T_i = 1150K$

$T_f = 450K$

$T_\infty = 325K$

$$Bi = \frac{h}{k}L = \frac{h}{k}\frac{V}{A} = \frac{h}{k}\frac{4/3\pi r^3}{4\pi r^2} = \frac{hr}{3k}$$

$$= \frac{25 \times 0.005}{3 \times 40} = 0.00104 \ll 0.1 \Rightarrow \text{Can use lumped model assumption}$$

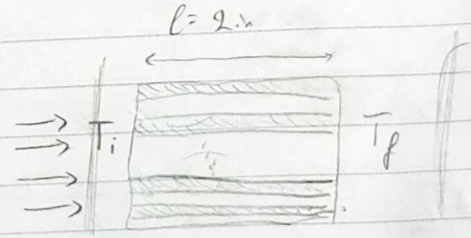
$$\Rightarrow \rho V c \frac{dT}{dt} = -hA(T - T_\infty)$$

$$\Rightarrow \frac{dT}{T - T_\infty} = \frac{-hA}{\rho V c}dt = \frac{-h\,4\pi r^2}{\rho\frac{4}{3}\pi r^3 c}dt = \frac{-3h}{\rho r c}dt$$

$$\Rightarrow \int_{T_i}^{T_f}\frac{dT}{T - T_\infty} = \int_0^t \frac{-3h}{\rho r c}dt \Rightarrow \ln\left(\frac{T_f - T_\infty}{T_i - T_\infty}\right) = \frac{-3h}{\rho r c}t$$

$$\Rightarrow \ln\left(\frac{450 - 325}{1150 - 325}\right) = \frac{-3 \times 25}{7800 \times 0.005 \times 600}t \Rightarrow \boxed{t = 588.8\ s}$$

Q.3:



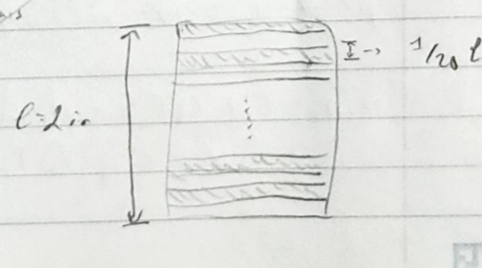
Parallel layers

heat rate = $A_\perp$ heat flux

$$\text{heat flux each layer} = \frac{-\Delta T}{l/k}$$

$$\Rightarrow \text{Total flux} = -\frac{10\Delta T}{l/k_{Cu}} - \frac{10\Delta T}{l/k_{glass}}$$

Perpendicular layers:

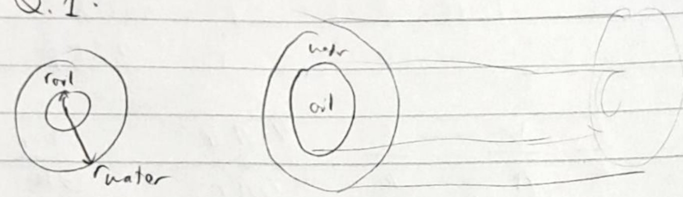


$$\text{heat flux total} = \frac{-\Delta T}{10\cdot\frac{l}{20k_{Cu}} + 10\cdot\frac{l}{20k_{glass}}}$$

$$= \frac{-\Delta T}{\frac{l}{2k_{Cu}} + \frac{l}{2k_{glass}}} = -\frac{2\Delta T}{l/k_{Cu} + l/k_{glass}}$$

$$\frac{q_{//}}{q_{\perp}} = \frac{A \cdot flux_{//}}{A\ flux_{\perp}} = 5$$

Q.1:



$\dot{V}_{oil} = 5.5\ gal/min$
$\dot{V}_{H_2O} = 11\ gal/min$
$T_{oil\,i} = 250°F$
$T_{oil\,f} = 120°F$
$T_{H_2O\,i} = 70°F$

$r_{oil} = 0.375\ in$
$r_{H_2O} = 0.75\ in$

$U_o = 110\ BTU/h\,°F\,ft^2$
$C_{H_2O} = 1\ BTU/lb\,°F$
$\rho_{H_2O} = 62.4\ lb/ft^3$

$C_{oil} = 0.55\ BTU/lb\,°F$
$\rho_{oil} = 52\ lb/ft^3$

Well insulated tube

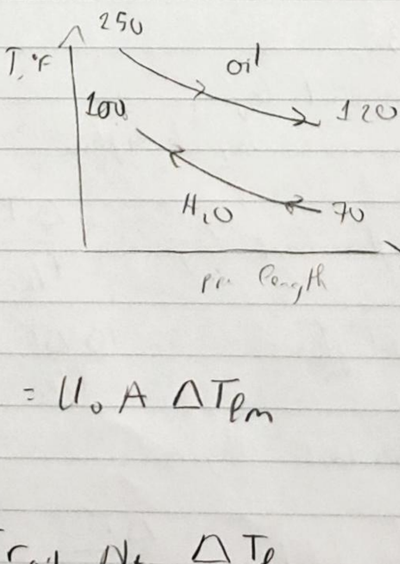
$\Rightarrow \dot{q}_{H_2O} + \dot{q}_{oil} = 0 \Rightarrow \dot{q}_{H_2O} = -\dot{q}_{oil} \Rightarrow \dot{m}_{H_2O} C_{H_2O} \Delta T_{H_2O} = -\dot{m}_{oil} C_{oil} \Delta T_{oil}$

$\Rightarrow \rho_{H_2O} \dot{V}_{H_2O} C_{H_2O} (T_{H_2O\,i} - T_{H_2O\,f}) = -\rho_{oil} \dot{V}_{oil} C_{oil} \Delta T_{oil}$

$\Rightarrow 62 \times 11 \times 1\ (70 - T_{H_2O\,f}) = -52 \times 5.5 \times 0.55\ (250 - 120)$

$\Rightarrow T_{H_2O\,f} = 100\ °F$

a) Counter current



$$\Delta T_{lm} = \frac{(250-100)-(120-70)}{\ln\left(\frac{250-100}{120-70}\right)}$$

$\Delta T_{lm} = 91.02\ °F$

for heat exchanger: $q = \dot{m} c \Delta T = U_o A \Delta T_{lm}$

$\Rightarrow \dot{m}_{oil} C_{oil} \Delta T_{oil} = U_o A \Delta T_{lm}$

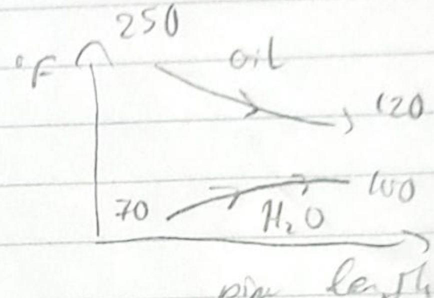
$\rho_{oil} \dot{V}_{oil} C_{oil} \Delta T_{oil} = U_o\ 2\pi r_{oil}\ N_t\ \Delta T_{lm}$

$N_t \equiv N^{\circ}$ of 12 ft tubing sections

$\dot{V}_{oil} = 5.5\ gal/min = 0.735\ ft^3/min = 44.1\ ft^3/h$

$\Rightarrow\ 52 \times 44.1 \times 0.55 \cdot (250 - 120) = 110 \cdot 2\pi \cdot \left(\frac{0.775}{12}\ in\right) 12 N_t \cdot 3102$

$\Rightarrow\ N_t = 6.95 = 7$ sections of tubing



b) Co-current flow

$$\Delta T_{lm} = \frac{(250 - 70) - (120 - 100)}{\ln\left(\frac{250 - 70}{100 - 100}\right)}$$

$$\Delta T_{lm} = 72.82\ ^{\circ}F$$

$\rho_{oil}\ \dot{V}_{oil}\ c_{oil}\ \Delta T_{oil} = U_o\ 2\pi r_{oil}\ 12 N_t\ \Delta T_{lm}$

$\Rightarrow\ N_t = 8.69 = 9$ sections of tubing